

## 4.1: Linear Inequalities in Two Variables

### Properties of Inequalities

**Substitution Property:** The inequality formed by substituting one expression for an equal expression is equivalent to the original inequality:

$$5x - 4x + 2 < 6$$

$$x < 4 \Rightarrow \text{The solution set is } \{x : x < 6\}$$

**Addition Property:** The inequality formed by adding the same quantity to both sides of an inequality is equivalent to the original inequality:

$$x - 4 < 6$$

$$x - 4 + 4 < 6 + 4$$

$$x < 10$$

$$x + 5 \geq 12$$

$$x + 5 + (-5) \geq 12 + (-5)$$

$$x \geq 7$$

**Multiplication Property** The inequality formed by multiplying both sides of an inequality by the same *positive* quantity is equivalent to the original inequality. The direction of the inequality is flipped when multiplying by a *negative* quantity:

$$\frac{1}{3}x > 6$$

$$3\left(\frac{1}{3}x\right) > 3(6)$$

$$x > 18$$

$$5x - 5 + 5 \leq 6x + 20 + 5$$

$$\frac{-x}{-1} \leq \frac{25}{-1}$$

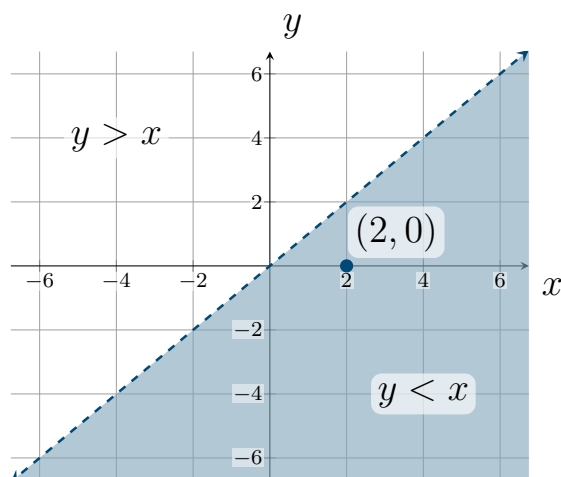
$$x \geq -25$$

## One Linear Inequality in Two Variables

### Definition.

Consider the inequality  $y < x$ :

The line created by this inequality divides the  $xy$ -plane into two **half-planes**. We can determine which half-plane is the solution region by selecting any point *not on the line* as a **test point**.



**Example.** Graph the inequality  $3x - 2y \leq 6$

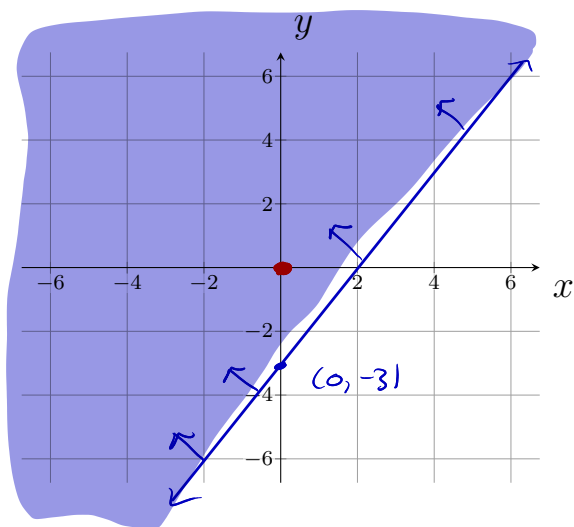
$$\begin{aligned} -3x + 3x - 2y &= 6 - 3x \\ \underline{-2y} &= \underline{-3x + 6} \\ -2 &\quad -2 \end{aligned}$$

$$y = \frac{3}{2}x - 3$$

Test point:

| $(x, y)$ | $3x - 2y \leq 6$  |
|----------|-------------------|
| $(0, 0)$ | $3(0) - 2(0) = 0$ |

Since  $0 \leq 6$ , we shade on the side of the test point

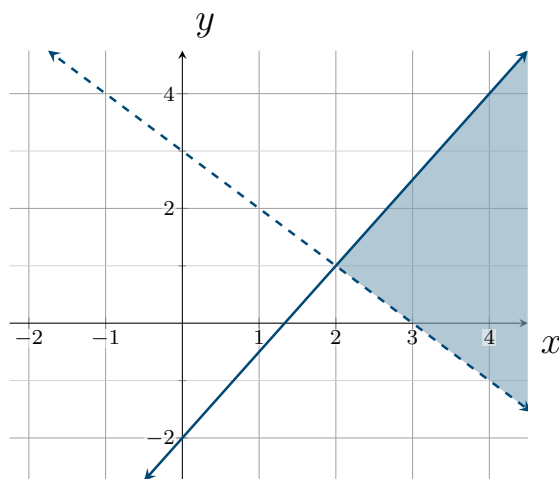


### Definition.

A **system of inequalities** has two or more inequalities in two or more variables. The solution of the system is the intersection of the individual solution sets.

### Example.

$$\begin{aligned} 3x - 2y &\geq 4 \\ x + y - 3 &> 0 \end{aligned}$$



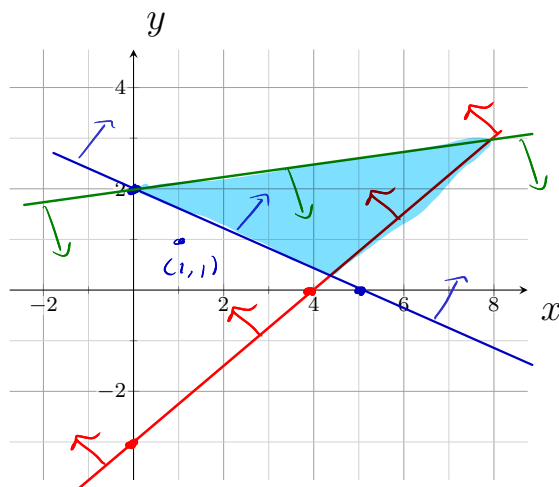
### Example. Graph the solution of the system

$$\begin{aligned} 3x - 4y &\leq 12 && (0, -3) \\ 2x + 5y &> 10 && (4, 0) \\ x - 8y &\geq -16 \end{aligned}$$

$$\begin{aligned} -x + x - 8y &\geq -16 - x \\ -8y &\geq -x - 16 \\ \frac{-8y}{-8} &\geq \frac{-x - 16}{-8} \\ y &\leq \frac{1}{8}x + 2 \end{aligned}$$

Test point:

| $(x, y)$ | $3x - 4y \leq 12$    | $2x + 5y > 10$      | $x - 8y \geq -16$ |
|----------|----------------------|---------------------|-------------------|
| $(1, 1)$ | $3(1) - 4(1) = -1$ ✓ | $2(1) + 5(1) = 7$ ✗ | $1 - 8(1) = -7$ ✓ |



**Example.** CDF Appliances has assembly plants in Atlanta and Fort Worth, where the company produces a variety of kitchen appliances, including a 12-cup coffee maker and a cappuccino machine. The following table shows each factory's assembly capabilities for the two products and the numbers needed to fill orders.

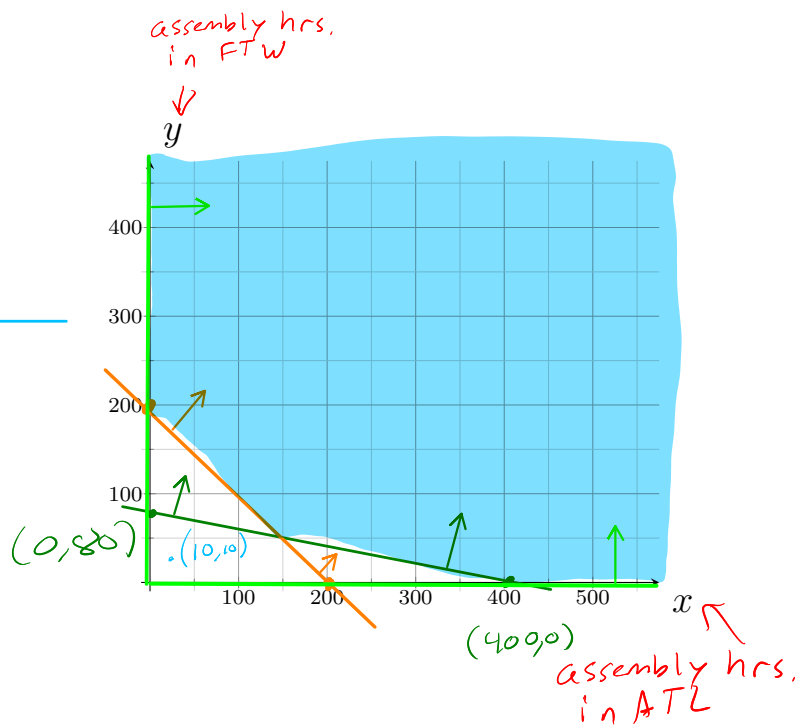
|                    | Atlanta | Fort Worth | Needed          |
|--------------------|---------|------------|-----------------|
| Coffee maker       | 160/hr  | 800/hr     | At least 64,000 |
| Cappuccino machine | 200/hr  | 200/hr     | At least 40,000 |

Write the system of inequalities that describes the number of assembly hours needed at each plant to fill the orders and graph the solution region for the system

$$\begin{aligned} \text{assembly hrs. in ATL} & \downarrow \\ \text{assembly hrs. in FTW} & \downarrow \\ \# \text{ coffee makers} & \rightarrow 160x + 800y \geq 64,000 \quad \leftarrow \begin{matrix} (0, 80) \\ (400, 0) \end{matrix} \\ \# \text{ cap. machines} & \rightarrow 200x + 200y \geq 40,000 \quad \leftarrow \begin{matrix} (0, 200) \\ (200, 0) \end{matrix} \\ \text{non-negative hours} & \rightarrow x \geq 0, y \geq 0 \end{aligned}$$

Test point:

| $(x, y)$  | $160x + 800y \geq 64,000$                         | $200x + 200y \geq 40,000$                         |
|-----------|---------------------------------------------------|---------------------------------------------------|
| $(0, 10)$ | $160(0) + 800(10)$<br>$= 1600 + 8000$<br>$= 9600$ | $200(0) + 200(10)$<br>$= 2000 + 2000$<br>$= 4000$ |



**Example.** A farm co-op has 6000 acres available to plant with  $\overset{x}{\text{corn}}$  and  $\overset{y}{\text{soybeans}}$ . Each acre of corn requires 9 gallons of fertilizer/herbicide and  $\frac{3}{4}$  hour of labor to harvest. Each acre of soybeans requires 3 gallons of fertilizer/herbicide and 1 hour of labor to harvest. The co-op has available at most 40,500 gallons of fertilizer/herbicide and at most 5250 hours of labor for harvesting. The number of acres of each crop is limited (constrained) by the available resources: land, fertilizer/herbicide, and labor for harvesting. Write the system of inequalities that describes the constraints and graph the solution region for the system.

fert. / herb.  $\rightarrow 9x + 3y \leq 40,500 \rightarrow y \leq -3x + 13,500$

labor  $\rightarrow \frac{3}{4}x + 1y \leq 5250 \rightarrow y \leq -\frac{3}{4}x + 5250$

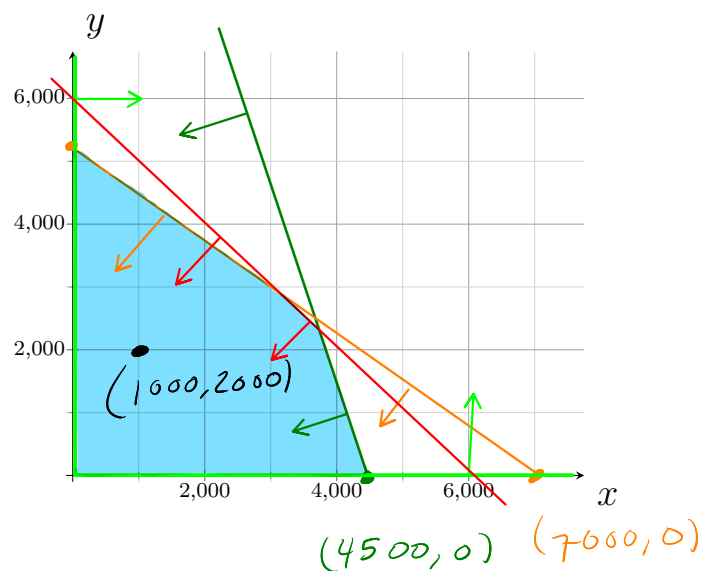
acres  $\rightarrow x + y \leq 6000 \rightarrow y \leq -x + 6000$

non-negative  $\rightarrow x \geq 0, y \geq 0$

# of crops

| $(x, y)$       | $9x + 3y \leq 40,500$<br>15,000                                | $\frac{3}{4}x + y \leq 5250$<br>2,750                                 |
|----------------|----------------------------------------------------------------|-----------------------------------------------------------------------|
| $(1000, 2000)$ | $9(1000) + 3(2000)$<br>$= 9000 + 6000$<br>$= 15000 \checkmark$ | $\frac{3}{4}(1000) + (2000)$<br>$= 750 + 2000$<br>$= 2750 \checkmark$ |

$x + y \leq 6000$   
 $1000 + 2000$   
 $= 3000 \checkmark$



**Example.** Graph the solution region for the system

$$\begin{aligned} 5x + 2y &\leq 54 && \begin{matrix} (0, 27) \\ (10.8, 0) \end{matrix} \\ 2x + 4y &\leq 60 && \begin{matrix} (0, 15) \\ (30, 0) \end{matrix} \\ x \geq 0, y &\geq 0 \end{aligned}$$

Then compute the corners of this region.

$$\begin{array}{r} 2(5x + 2y = 54) \\ - 1(2x + 4y = 60) \\ \hline 8x + 0y = 48 \\ \underline{8} \quad \quad \quad \underline{8} \\ x = 6 \end{array}$$

$$\begin{aligned} 2(6) + 4y &= 60 \\ 12 + 4y &= 60 \\ 4y &= 48 \\ y &= 12 \end{aligned}$$

Test point:

| $(x, y)$   | $5x + 2y \leq 54$ <span style="color: green;">120</span>                      | $2x + 4y \leq 60$ <span style="color: orange;">80</span>                    |
|------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| $(20, 10)$ | $5(20) + 2(10)$<br>$= 100 + 20$<br>$= 120$ <span style="color: red;">✗</span> | $2(20) + 4(10)$<br>$= 40 + 40$<br>$= 80$ <span style="color: red;">✗</span> |

Corners:

$(0, 0)$   
 $(10.8, 0)$   
 $(6, 12)$   
 $(0, 15)$

